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Advanced modeling options

This chapter describes the advanced modeling options in Eclipse:

- [Run Eclipse in coupled geomechanics mode](#)
- [CO2 storage](#)

Run Eclipse in coupled geomechanics mode

Petrel Reservoir Geomechanics (Petrel RG) is a Petrel module which works with the Visage geomechanics simulator and the Eclipse reservoir simulator to provide 3D and 4D modeling of rock stresses and deformations in and around a reservoir. This helps to predict where subsidence and compaction might occur and improves the modeling of fluid flow based on these geomechanical changes. Using Petrel RG, you create a mechanical earth model that describes the elastic, plastic and strength behavior of the reservoir and its surrounding formations. Using these properties along with information on changes in pore pressure, temperature and saturation/composition from Eclipse, Visage calculates the reservoir stresses and strains and modifies pore volume and/or permeability as a result.

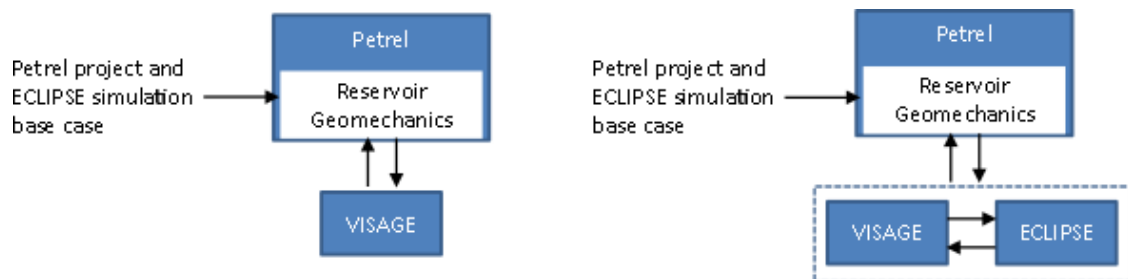


Figure 14.1. Petrel Reservoir Geomechanics

Using the Petrel RG workflow the two main methods of performing the geomechanics calculations are shown in the figure: one-way coupling to the left and two-way coupling to the right.

- With one-way coupling Visage uses the information in a Petrel project and the results and properties from an Eclipse simulation, stored in the Eclipse restart files, to calculate changes in the reservoir stresses and strains due to the changes in pressure, temperature and saturation. There is no transfer of data or results from Visage to Eclipse.

With two-way coupling, you can choose explicit or implicit coupling:

- Explicit coupling

At certain times in the simulation, known as coupled timesteps, Petrel RG stops the Eclipse run and processes results from the restart files. During a two-way coupled simulation, Petrel RG restarts the Eclipse run with changes in pore volume and/or permeability calculated based on the changes in stresses and strains.

- Implicit coupling

Petrel RG stops the Eclipse run at a coupled timestep, and processes results from the restart files. It then restarts Eclipse from the previous timestep with modified pore volume and/or permeability. This process is repeated until the change in pore volume calculated by Eclipse over the period agrees with that calculated by Visage.

Using the Petrel RG workflow, you do not have to know about, or manually set, any Visage or Eclipse keywords. You control the simulation and analyze the results through the Petrel RG interface. For information, see the *Reservoir Geomechanics* help in Petrel.

The alternative to this workflow is to use Eclipse ["Rock compaction"](#). This provides a simplified and less rigorous analysis that can account for some basic geomechanical effects, where simple elastic behavior is considered in the reservoir only.

Using Petrel RG with Visage with Eclipse

Note: The interactions between the Visage and Eclipse simulators are controlled using the Petrel Reservoir Geomechanics (Petrel RG) workflow. The interactions described in this section are not visible during the simulation and are provided for reference only. The Petrel RG simulation logs contain records of the keyword data written into the Eclipse files during the geomechanics simulation.

During the workflow, Petrel RG writes keywords into the Eclipse files to control the interaction of the simulators:

- `VISOPTS` in the `RUNSPEC` section to activate the workflow
- `VISDATES` and `VISOPTS` in the `SOLUTION` section to specify the coupled timesteps dates and the simulation options.

You do not have to set any of these keywords by hand, or edit any simulation files as these interactions occur according to settings you specify in the Petrel RG workflow. This section provides an overview of the simulator interactions.

1. Petrel RG activates the coupling of simulators with the `VISAGE` keyword.
2. Petrel RG specifies the list of report dates at which geomechanics calculations are required using the `VISDATES` keyword.

To request the available report dates from an Eclipse simulation run, Petrel RG runs in initialization mode by setting item 1 of the `VISOPTS` keyword to `YES`. In initialization mode, Eclipse writes out a file containing information about all report steps that would be performed in the simulation (the model is run as though `NOSIM` is present). This file has the same root name as the simulation, but with an `EDESC` extension (for example, when running a simulation model `NAME.DATA`, this file will be `NAME.EDESC`). Also in initialization mode, all requests for couple timesteps are ignored; and `GRID`, `INIT` and time zero restart files are output.

3. Petrel RG can set Eclipse to exit at a coupled timestep (once the restart files have been written) by setting item 2 of the `VISOPTS` keyword to `YES`. This feature is turned off by default but Petrel RG turns it on for explicit or implicit coupling in the geomechanics workflow.
4. When the field pressure at a report time has changed by more than the relative tolerance (specified in item 4 of the `VISOPTS` keyword) since the last coupled timestep, an additional step will be performed at a timestep called a dynamic stress step. Note that this feature must be turned off when repeating the calculations over a stress step interval in the implicit coupling, and any dynamic stress steps which have been performed must be added to the `VISDATES` list. The dynamic stress step mechanism is turned off by default, but Petrel RG can turn it on by setting item 3 of the `VISOPTS` keyword to `YES`.

Note: Petrel RG can handle a maximum of 9999 stress steps in its workflow. If you request more than 9999 stress steps in the `VISDATES` keyword, or if a dynamic stress step would be triggered after 9999 stress steps have already occurred, then Eclipse will exit with an error.

5. In order to guarantee that restart files are generated at all of the coupled timesteps, any `FREQ` mnemonic specification in the `RPTRST` keyword is ignored and `FREQ=1` (that is, output a restart file at every report step) is enforced by default. If you want a smaller amount of data is required in the restart files, you can reduce the frequency of output by:

- a. specifying the `RPTRST BASIC` mnemonic as 3, 4 or 5,

- b. setting the `FREQ` mnemonic to the desired output frequency and
- c. setting item 6 of the `VISOPTS` keyword to `YES`.

Note: When modifying the restart output frequency, you must ensure that there will still be restart information available at all required stress step dates.

6. In the geomechanics workflow, the minimum requirement for restart file content is that it can be used to restart the run. For this reason, the `RPTRST BASIC=2` mnemonic is enforced when the `VISAGE` keyword is present. In addition, the `RPTRST ROCKC` mnemonic is enforced so that the modified pore volume and transmissibility multiplier are present in the restart files. By default, any other property requests will be ignored. If additional properties are required in the restart file, these must be specified in the `RPTRST` keyword, and item 1 of the `VISOPTS` keyword must be set to `YES`.
7. Whenever a new coupled timestep is encountered, Eclipse writes a list of all coupled timesteps to a small file with the same root name as the simulation, but with a `VDESC` extension (for example, when running a simulation model `NAME.DATA`, this file will be `NAME.VDESC`). This information allows Petrel RG to know which of the requested coupled timesteps have been performed, and also whether any dynamic coupled timesteps have been performed.

At the beginning of the workflow, this `VDESC` file needs to be emptied (if it already exists from a previous run). Eclipse is instructed to empty this file during Petrel RG initialization, which Petrel RG requests by setting item 1 of the `VISOPTS` keyword to `YES`.

CO₂ storage

	Eclipse 100
x	Eclipse 300

In Eclipse 300 there are several options available in order to study the storage of CO₂ under various conditions. This chapter describes the [CO2STORE](#) option, for CO₂ - H₂O systems with salts. In addition the following options are available.

- The [CO2SOL](#) option, typically used to study CO₂ storage in oil reservoirs in isothermal mode - water is not allowed in the oil/gas phase.
- The [GASWAT](#) option, general equation of state for gas-water systems in isothermal mode.
- The [COAL](#) option for coal bed methane reservoirs.

Since the [CO2STORE](#), [CO2SOL](#) and the [GASWAT](#) options estimate the fluid properties differently, they cannot be used together.

In thermal mode the [CO2STORE](#) option can be used with any number of components and can include a liquid oil phase as described in [Thermal CO2STORE](#).

Thus, in isothermal mode, [CO2STORE](#) can be used for CO₂ storage simulations in saline aquifers, [GASWAT](#) can be used for CO₂ storage simulations in depleted gas reservoirs, and [CO2SOL](#) can be used for CO₂ storage simulations in depleted oil reservoirs. However, in thermal mode, [CO2STORE](#) thermal can be used to model CO₂ storage simulations in saline aquifer and depleted oil and gas reservoirs. [GASWAT](#) and [CO2SOL](#) options are not compatible with thermal mode.

Mode	CO ₂ storage in saline aquifers	CO ₂ storage in depleted gas reservoirs	CO ₂ storage in depleted oil reservoirs
Isothermal mode	CO2STORE option	GASWAT option	CO2SOL option
Thermal mode	CO2STORE option	CO2STORE option	CO2STORE option

The [CO2STORE](#) option

For the [CO2STORE](#) option three phases are considered, a CO₂ rich phase, an H₂O rich phase and a solid phase. The CO₂ rich phase is labeled the gas phase while the H₂O rich phase is labeled the aqueous phase (liquid phase). The mutual solubilities of CO₂ and H₂O are calculated to match experimental data for CO₂ - H₂O systems under typical CO₂ storage conditions: 12-100 °C and up to 600 bar. The model is also applicable at elevated temperatures as detailed in [Spycher and Pruess \(2009\)](#), up to 250 °C. Fluid components are specified with the [COMPS](#) keyword. Allowed component names are currently CO₂, H₂O and the salts [NACL](#), [CACL2](#) and [CACO3](#).

Component name	Phases
CO ₂	Aqueous/Gas
H ₂ O	Aqueous/Gas
NACL	Aqueous/Solid
CACL2	Aqueous/Solid
CACO3	Aqueous/Solid

Table 14.1: Components that can be used with the [CO2STORE](#) option

The option is similar to the [GASWAT](#) option, but the salt components of the aqueous or solid phase are tracked as part of the fluid system. Also the partitioning of the component H₂O in the gas phase has been matched accurately. Fluid properties are assigned internally based on the names; this means that a minimum amount of data is needed in order to supply the fluid properties.

Phase splitting

The partitioning of CO₂ and H₂O in the aqueous and gas phase follows the procedure given by Spycher and Pruess (2005 and 2009). The salts are assumed to stay in the aqueous phase unless the [SOLID](#) option is activated.

With the [SOLID](#) option the components NaCl, CaCl₂ and CaCO₃ can be present in both the aqueous phase and the solid phase according to the following equilibrium reactions:

- $\text{H}_2\text{O} \rightleftharpoons \text{H}^+ + \text{OH}^-$
- $\text{CO}_2 + \text{H}_2\text{O} \rightleftharpoons \text{HCO}_3^- + \text{H}^+$
- $\text{HCO}_3^- \rightleftharpoons \text{CO}_3^{2-} + \text{H}^+$
- $\text{CaCl}^+ \rightleftharpoons \text{Ca}^{2+} + \text{Cl}^-$
- $\text{NaCl(s)} \rightleftharpoons \text{Na}^+ + \text{Cl}^-$
- $\text{CaCl}_2\text{(s)} \rightleftharpoons \text{CaCl}^+ + \text{Cl}^-$
- $\text{CaCO}_3\text{(s)} \rightleftharpoons \text{Ca}^{2+} + \text{CO}_3^{2-}$

If only NaCl is present a simplified procedure can be used and the maximum NaCl solubility in the aqueous phase is calculated by [Potter II et al \(1977\)](#).

$$WtNaCl = 26.218 + 0.0072T + 0.000106T^2$$

where T is the temperature in °C.

It is recommended that the [SOLIDMMS](#) keyword is also used in order to model the decrease in mobility as a function of the solid saturations. The gas and aqueous saturations are normalized to be between zero and unity, so the two phase water and gas relative permeability curves can be used. However it should be noted that the effective fluid volume is given by

$$FV = (1-S)PV$$

where PV is the pore volume and S is the solid saturation. It is possible to specify a target solid saturation change for each timestep by using item 21 of [TSCRIT](#). A small value will usually enhance the stability.

Activity coefficient models

The CO₂ and H₂O activity are computed following [Spycher and Pruess \(2005\)](#) and [Spycher and Pruess \(2009\)](#). This is done using Margules expressions:

$$\ln(\gamma_{\text{H}_2\text{O}}) = (A_M - 2A_M x_{\text{H}_2\text{O}})x_{\text{CO}_2}^2$$

$$\ln(\gamma_{\text{CO}_2}) = 2A_M x_{\text{CO}_2} x_{\text{H}_2\text{O}}^2 e^{(\gamma_{\text{CO}_2})'}$$

where:

$$A_M = 0 \quad (T \leq 100^\circ\text{C})$$

$$A_M = a \cdot (T-100) + b \cdot (T-100)^2 \quad (T > 100^\circ\text{C})$$

In these expressions, a and b are regressed parameters and x_{CO_2} and $x_{\text{H}_2\text{O}}$ denote the aqueous phase mole fractions. The CO_2 activity coefficient is modified for salting out effects by γ_{CO_2}' .

There are two models that can be applied in order to compute γ_{CO_2}' :

- The Rumpf et. al 1994 model as detailed in [Spycher and Pruess \(2005\)](#), applicable up to 160 °C.
- The model given in [Spycher and Pruess \(2009\)](#), applicable up to 250 °C.

To choose the model, use the keyword `ACTCO2S`.

Density

The gas density is obtained by a cubic equation of state tuned to accurately give the density of the compressed gas phase, following [Spycher and Pruess \(2005\)](#). A modified Redlich-Kwong equation of state is used, where the attraction parameter is made temperature dependent:

$$P = \left(\frac{RT_K}{V - b_{\text{mix}}} \right) - \left(\frac{a_{\text{mix}}}{T_K^{1/2} V(V + b_{\text{mix}})} \right)$$

where V is the molar volume, P is the pressure, T_K the temperature in Kelvin, R is the universal gas constant and a_{mix} and b_{mix} are the attraction and repulsion parameters. For more information see also [Spycher and Pruess \(2009\)](#).

The transition between liquid CO_2 and gaseous CO_2 will lead to rapid density changes for the CO_2 -rich phase. The simulator uses a narrow transition interval between the liquid state and gaseous state.

For the brine density the simulator first computes the density of pure water. The default model uses an analytical form of the water density as detailed by *The International Association for the Properties of Water and Steam* (IAPWS-IF97). In order to use the pre-2010.1 water density calculations as detailed in [Kell and Whalley \(1975\)](#), item 232 in `OPTIONS3` can be put to unity. Ezrokhi's method is then applied to calculate the effect of salt and CO_2 as described in [Zaytsev and Aseyev \(1993\)](#), equation 6.69.

Viscosity

The CO_2 gas viscosity is calculated from [Vesovic et al \(1990\)](#) and [Fenghour et al \(1998\)](#) while the aqueous viscosity is calculated using Ezrokhi's method, equation 6.71.

Aqueous speciation

It is possible to output the different ions and chemical components in the aqueous phase by adding AQSP to the `RPTRST`, `RPTSOL`, or `RPTSCHEd` keyword. Similarly the pH is output by adding AQPH. Additionally, the SUMMARY keywords BAQSP and BAQPH can be used.

Names of the species considered are given below

Aqueous species	
aH2O	Pure water
aCO2	Carbon dioxide
aH+	Hydrogen ion

Aqueous species	
aOH-	Hydroxide ion
aCO3-2	Carbonate ion
aHCO3-1	Bicarbonate ion
aCl-	Chloride ion
aNa+	Sodium ion
aCa++	Calcium ion
aCaCl+	Calcium monochloride ion

Table 14.2: Aqueous Species

Note: As an example, the total amount of aqueous NaCl is given by the molar density, named MLSC followed by the component index, while the speciation outputs the ions a NA⁺ and a CL⁻.

pH calculation

The pH is calculated as

$$pH = -\log(\zeta_H \cdot m_H)$$

where

ζ_H is the activity coefficient of the H⁺ ion and

m_H is the molality of the H⁺ ion.

Analytic water aquifers

An analytic water aquifer defined by [AQUCHWAT](#), [AQUCT](#), [AQUFETP](#) or [AQUFLUX](#) contains only the “H2O” component. It is possible to modify the aquifer composition using the [AQSTREAM](#) keyword. Any component is allowed to leave or enter the reservoir. Additionally, the aquifer water properties are calculated as described above.

If [item 131](#) of the [OPTIONS3](#) keyword is set to 1, this reverts to the pre-2007.2 behavior where the water entering / leaving the reservoir is made up entirely of the “H₂O” component. If [item 119](#) of [OPTIONS3](#) is set to 1, this reverts to the pre-2007.1 behavior where the aquifer water properties were calculated using the water PVT tables. See ["Aquifer modeling facilities"](#).

Using the CO₂ storage model

RUNSPEC section

The option is invoked by the [CO2STORE](#) keyword. If a solid phase is considered the [SOLID](#) keyword must be specified. The capillary pressure mode is selected by default so that the capillary pressure is added to the gas phase pressure. To subtract the capillary pressure from the aqueous phase pressure use [item 8](#) of the [OPTIONS3](#) keyword. To run with a saturation weighted capillary pressure option the keyword [FORMOPTS](#) can be used with the mnemonic [PCSPLIT](#). For runs with large capillary pressure it is recommended to set item 8 of [OPTIONS3](#), so the system pressure is the gas phase pressure. Also note that the [WATER](#) keyword should not be used with the [CO2STORE](#) option.

PROPS section

The fluid components are set by specifying the names in the **COMPS** keyword.

The water relative permeability is specified using the **WSF** keyword, while the gas relative permeability and the gas-water capillary pressure are entered using the **GSF** keyword.

If only a single phase is present the composition may be entered using either **ZI** or with respect to depth **ZMFVD**.

If an initial gas-water contact exists within the reservoir you may specify either the vapor or the aqueous phase composition at the contact (**ZMFVD** or **COMPVD** may be used).

To enable molecular diffusion, the keywords **DIFFCWAT** and **DIFFCGAS** should be used. Cross phase diffusion can be included by the keywords **DIFFCWG** and **DIFFCGW**. Note that without the cross phase diffusion option no diffusive flow is computed between two neighbor cells that are single phase, one being aqueous and the other gas. The interaction with the formation water could therefore be slightly under predicted without including the cross phase diffusion term. For more information see "[Diffusion](#)".

SOLID properties need to be assigned in the usual way. The component reference density is assigned using the keyword **SDREF**. The **SOLIDMMS** keyword can be used to specify mobility multipliers.

For temperatures above 160 °C an alternative activity coefficient model for the aqueous CO₂ should be considered using the keyword **ACTCO2S**.

SUMMARY section

The keywords **FWCD**, **FGCDI** and **FGCDM** can be used to output the field total molar amount of CO₂. **RWCD**, **RGCDI** and **RGCDM** can be used to output the same quantities on a regional basis.

Field	Region	Description
FWCD	RWCD	Dissolved in water
FGCDI	RGCDI	In an immobile gas phase
FGCDM	RGCDM	In a mobile gas phase

Notes

- Quantities such as **FGIP**, **FGIPL** and **FGIPG** depend on the state of the CO₂ at standard conditions. This means that if all the CO₂ injected is in a dissolved state these quantities are zero. Early in a simulation, **FGIP** might be zero (because at standard conditions, all the CO₂ injected could be dissolved in the amount of water in the field) while **FGIPG** is non-zero (because there is a high concentration of CO₂ close to the injector and it cannot all dissolve in its current location). The field molar amount injected for a given component can be obtained using the keywords **FCMIT** and **FCMIP**. The quantity which corresponds to the PRT report of CO₂ dissolved in water is **FWCD**.
- Also the aqueous phase is considered as “hydrocarbon” when calculating the “hydrocarbon weighted” field pressure reported by **FPR** and **FPRH**. For a comparison with a simulation involving an oil phase such as **CO2SOL**, the “pore volume weighted” pressure returned by **FPRP** can be used. Additionally, setting [item 151](#) of the **OPTIONS3** keyword to 1 ensures that only the gas phase is treated as “hydrocarbon”.

SOLUTION section

The `EQUIL` keyword is used for equilibration. An initial solid saturation and corresponding composition can be assigned by the keywords `SSOLID` and `SMF`. Some care must be taken as this might not be in equilibrium. The initial flash will however put the minerals in equilibrium with the formation water.

SCHEDULE section

To specify a CO₂ injector the keyword `WINJGAS` can be used to define the composition of the injector. The composition of a water injector can also be specified by using this keyword. However, to specify the composition of an injected water stream, the keywords `WELLSTRW` and `WINJW` are available with the `CO2STORE` option. When using the `SOLID` option it is possible to specify a target solid saturation change by using item 21 of `TSCRIT`.

If the adaptive implicit solution technique (the `AIM` option) is used, the `AIMCON` keyword can be used in order to increase the number of implicit cells allowing large stable timesteps. Typically this will be the case at the end of a simulation when CO₂ injection has stopped.

Compatibility

The `CO2STORE` option is currently not compatible with the multisegment well option.

Note: Salt precipitation is enabled by the use of the `SOLID` option. If the `SOLID` option is not used, salts are assumed to stay in the liquid phase. For such cases some residual water might become more and more concentrated, with salt resulting in unphysically dense water. However, this water will typically become immobile, so the overall simulation is only marginally affected. Typically this will only occur near a CO₂ injection well, due to injection of dry CO₂. The model applies a maximum NaCl concentration for the CO₂ solubility model, corresponding to the value of salting out while other salts are limited to a fixed value of 5 Molar in the CO₂ solubility model

Thermal CO2STORE

The `CO2STORE` option can be used with the `THERMAL` option for live oil and gas. The fluid properties are defined similarly to a standard thermal case, but the CO₂ component is allowed to dissolve in the aqueous phase. When using the `CO2STORE` option together with `THERMAL` it is assumed that the temperature range of the reservoir is limited to 12-250 °C in a model with additional hydrocarbons. For models with no additional hydrocarbons, the range is 1-250 °C.

If no oil K-values are supplied the phases assumed is as for a standard `CO2STORE` case, a aqueous water rich phase and a CO₂ rich gas phase. The density of the CO₂ rich phase and the water vapor is computed as for the standard `CO2STORE` option. Additional components are taken as water components unless you supply water K-values for these components or define them as gas only components. This is done by the keywords `KWTABTn` and `CVTYPE`. Note that the CO₂ and H₂O component is allowed to be defaulted when using the `CO2STORE` option. If this is done the usual computations for the solubilities of CO₂ and H₂O will be applied.

Additional oil/gas components can be made non-soluble in water by defaulting them in the `KWTABTn` keyword. Their viscosities can be defined in the `GASVISCT` and `OILVISCT` keywords, and if the viscosities of CO₂ and H₂O are defaulted, the usual computations for their viscosities will be applied.

In `THERMAL` it is possible to define the H₂O components as part of the components defined in `CNAMES`, or if left out it will be added as the last component of the system. If the H₂O component is specified it is also

possible for H₂O to be in the oil phase. However this is only allowed if oil K-values are supplied using [KVTA_BT_n](#), where you need to supply the split of H₂O between oil and vapor.

Note: With thermal CO₂STORE, the aqueous mole fractions are named [AMF](#), while [XMF](#) refers to the oil phase mole fractions.

Note: If the only explicitly supplied component is CO₂, the C_{NAMES} keyword is not required.

Density and viscosity

The density of the aqueous phase is computed as described by the keyword [DENAQA](#), equation 6.69) and [VISCAQA](#), equation 6.71. Default values will be assigned for CO₂/CH₄, NaCl/CaCl₂/CaCO₃ component. The vapor phase density is computed as for a standard thermal case, but the CO₂ and H₂O contribution to the phase specific volume is computed internally.

Similarly for vapor phase viscosity the CO₂ and H₂O contribution when calculating the phase viscosity are computed internally unless a component specific value for the viscosity is assigned.

Enthalpy

The aqueous enthalpy is computed by the parameters given in the keywords [SPECHW1](#) and [SPECHW2](#).

For the gas phase the usual parameters given for a thermal live oil run should be supplied, note that for CO₂ the parameters defining the enthalpies for both an oil and a vapor should be given even if not an oil phase is considered.

The H₂O molar enthalpy in the vapor phase is computed internally.

An alternative enthalpy formulation for all phases is available for systems with CO₂ and H₂O without additional hydrocarbon, which takes the CO₂ enthalpy from internal tables generated with the Symmetry* process software platform. These tables contain enthalpy as a function of pressure, temperature, and (optionally) CO₂ molar fraction of the phase, up to a pressure of 600 bar and a temperature of 250°C. Within the table, values are linearly interpolated. Above these values, extrapolation is linear for pressure and constant for temperature.

This option is controlled using the [CO2ENTH](#) keyword. When it is activated, the [SPECH_x](#), [HEATVAP_x](#) and [ZFACT1](#) keywords are not required. Zero enthalpy is taken to be the enthalpy of the liquid phase at the temperature and pressure specified in the [STCOND](#) keyword.

Relative permeability

The relative permeability needs to be of the three phase type. If an oil phase is not considered, the oil relative permeability needs to be assigned dummy values.

Note: [WSF](#) and [GSF](#) cannot be used to specify the relative permeabilities.

Low temperature option

The Spycher-Pruess ([Spycher and Pruess \(2005\)](#) and [Spycher and Pruess \(2009\)](#)) method is normally limited to a temperature range of 12-250 °C. For systems with CO₂ and H₂O without additional hydrocarbon, lower temperatures are allowed and calculations are based on a linear extrapolation of CO₂ and H₂O k-values from 12 °C downward to 1 °C. At these low temperatures, both component k-

values are approximately linear with temperature. This option can be disabled by setting [item 438](#) of the `OPTIONS3` keyword to 1.

Detection of hydrate formation

An additional `CO2STORE` option exists to detect the formation of $\text{CO}_2/\text{H}_2\text{O}$ hydrates at low temperatures. It is only applicable to systems with CO_2 and H_2O without additional hydrocarbon. Several salts can be modeled using the `CO2STORE` option. These have the effect of depressing temperatures at which hydrates can form in this fluid system or, in other words, are hydrate inhibitors. A hydrate formation envelope was fitted as a function of pressure and salt concentration from predictions with the Multiflash hydrate calculator ([Infochem, 2019](#)). The effect of all salts is modeled using an equivalent set of NaCl ions in solution. The accuracy of these envelope predictions is $\pm 2^\circ$ and the accuracy of the fit is $\pm 1.2^\circ$. Two methods are available to report hydrate formation:

- A summary vector can be defined that reports hydrate formation in grid blocks using the summary keyword `BHYD` (or `LBHYD` for local grid blocks).
- A 3D array can be requested where a value of 1 indicates hydrate formation, invoked with the `HYDRATE` mnemonic in the [RPTRST](#) or [RPTSCHED](#) keyword.

Initializing the study

The typical conditions is that the reservoir is water. It is possible to assign the initial aqueous mole fractions by the [AMFVD](#) keyword when using the [EQUIL](#) keyword. If the reservoir is initialized by enumeration the aqua mole fractions are assigned by the [AMF](#) keyword.

Compatibility

The thermal [CO2STORE](#) option is currently not compatible with the multisegment well option.